



A Hybrid Machine Learning model for Accurate Autism Diagnosis

Abstract

Autism Spectrum Disorder (ASD) is a developmental disorder that affects communication, behaviour, and social interaction. Early detection of autism is very important because it helps in providing timely support and treatment for individuals affected by the disorder.

The traditional autism diagnosis system has several limitations. The diagnosis process is time consuming and often requires multiple visits to healthcare specialists. It also depends heavily on expert judgment, which may sometimes lead to subjective interpretations. In many regions, access to trained professionals is limited, causing delays in early detection. Additionally, traditional methods may not efficiently analyse large amounts of behavioural or medical data, which can reduce the accuracy of diagnosis. These limitations highlight the need for automated and intelligent systems to support autism detection.

The proposed system introduces a Hybrid Machine Learning Model for accurate autism diagnosis. This system uses multiple machine learning algorithms combined together to improve prediction performance and accuracy. The model analyses behavioural, screening, and demographic data to identify patterns related to Autism Spectrum Disorder. By training the hybrid model on historical datasets, the system can predict the likelihood of autism in individuals with higher accuracy. The proposed system aims to assist healthcare professionals by providing a fast, reliable, and data-driven decision support tool for early autism diagnosis.

Objectives

The objectives of the proposed system are:

- To develop a hybrid machine learning model for accurate autism diagnosis.
- To analyze behavioral and screening data using machine learning techniques.
- To improve prediction accuracy by combining multiple machine learning algorithms.
- To support early detection of Autism Spectrum Disorder.
- To assist medical professionals with an automated diagnostic support system.



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System Requirements

a. Software Requirements :

- Operating System: Windows 10 / Windows 11
- Programming Language: Python and its Libraries

b. Hardware Requirements :

- Processor: Intel Core i5
- RAM: Minimum 8 GB
- HDD: 1TB

Mini Project Batch No.: A19

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